

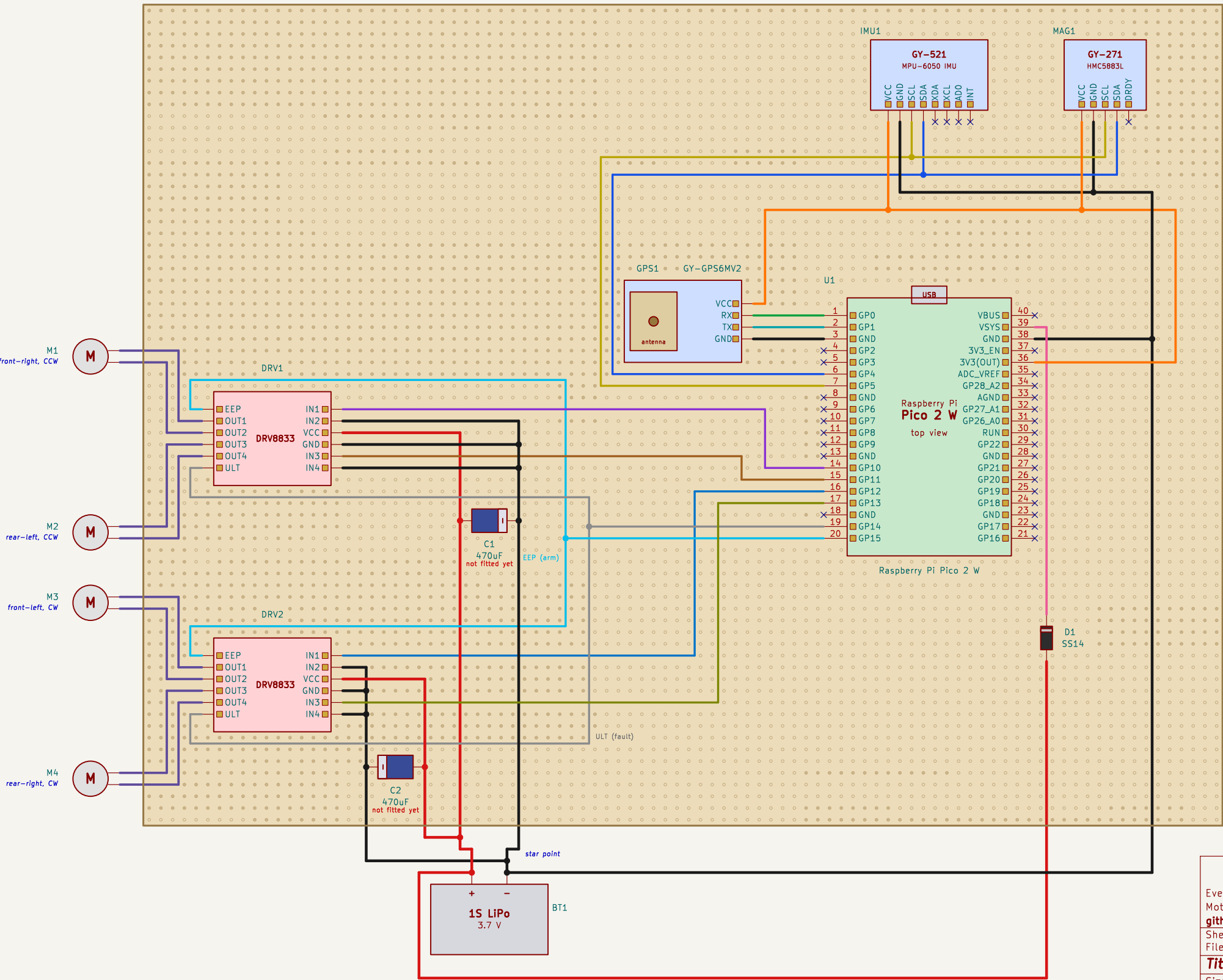
PICO DRONE – PROTOBOARD WIRING DIAGRAM

Every line is a real wire. A dot means wires are joined there; lines that cross without a dot are NOT joined. Motors and battery sit off the board on their own leads.

WIRE COLOURS

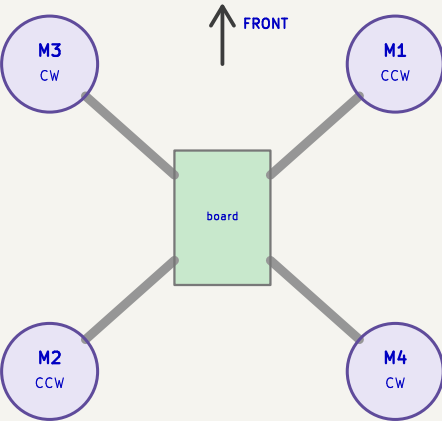
- Battery + (to DRV VCC, D1)
- Ground (star at battery -)
- VSYS (D1 -> Pico pin 39)
- 3.3 V (Pico pin 36 -> sensors)
- Motor leads
- I2C SDA (GP4, pin 6)
- I2C SCL (GP5, pin 7)
- GP0 (pin 1) -> GPS RX
- GPS TX -> GP1 (pin 2)
- Arm / EEP (GP15, pin 20)
- M1 PWM (GP10, pin 14)
- M2 PWM (GP11, pin 15)
- M3 PWM (GP12, pin 16)
- M4 PWM (GP13, pin 17)
- Fault / ULT (GP14, pin 19)

PROTOBOARD



WHERE THE MOTORS GO

top view, nose up



DRV1 drives M1 + M2, DRV2 drives M3 + M4
(each pair is diagonal)

BUILD NOTES

- Match each wire by the LABEL printed on the board. DRV8833 sellers order their pins differently.
- Pico numbers 1–40 are physical pins, USB at top. Pins 3 and 38 are both GND inside the Pico.
- Battery (-) is the star point: separate black wires to DRV1, DRV2 and the Pico GND trunk. The Pico only gets it through D1 (stripe up).
- Battery (+) goes straight to both DRV VCC pins. The Pico gets it through D1 (stripe up).
- C1, C2 = 470 uF, stripe (-) to GND, legs right at the driver. Required – not fitted yet.
- IN2 + IN4 tied to GND = STANDARD wiring (set lift.py WIRING = "STANDARD").
- Spin direction = which lead goes to which OUT. Swap a motor's two leads to reverse it.
- Connect the battery before USB until C1/C2 are in.

Every connection drawn as a wire, boards in physical pin order
Motor wiring: STANDARD
github.com/krishaanth5831/pico-drone
Sheet: /
File: pico-drone-wiring.kicad_sch

Title: Pico Drone – protoboard wiring

Size: A3
KiCad E.D.A. 9.0.8

Date: 2026-10-01

Rev: 1

Id: 1/1